Chapter 13. Bilinear Forms

Quadratic forms, orthogonal and symplectic groups, duality

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Armed with groups, quotient groups and quotient spaces, we return to inner product spaces, aiming at $\mathbf{O}(n)$ and $\mathbf{U}(n)$. To study inner product spaces properly one needs bilinear and Hermitian forms, and for these the dual space is indispensable. From now on all vector spaces are finite-dimensional.

13.1 Bilinear forms

Motivation. Dropping positive definiteness from a real inner product leaves a bilinear form. Such a form is completely determined by its values on pairs of basis vectors, i.e. by a matrix: "bilinear forms are matrices", just as linear maps are.

For a bilinear form $B : V \times V \to F$, a basis $\mathfrak{B} = \{v_i\}$ of V, $v = \sum a_i v_i$, $w = \sum b_i v_i$:

$$B(v,w) = B\Big(\sum_i a_i v_i, \sum_j b_j v_j\Big) = \sum_{i,j} a_i b_j B(v_i, v_j). \tag{$*$}$$

Observation 13.1.1

With $[B]_{\mathfrak{B}} = (B(v_i,v_j))_{n\times n}$, $B(v,w) = [v]_{\mathfrak{B}}^t\,[B]_{\mathfrak{B}}\,[w]_{\mathfrak{B}}$ for all $v, w \in V$. $[B]_{\mathfrak{B}}$ is the **matrix of** B with respect to $\mathfrak{B}$.

Proof. $(*)$ reads $\sum_{i,j} a_i B(v_i,v_j) b_j = [v]_{\mathfrak{B}}^t [B]_{\mathfrak{B}} [w]_{\mathfrak{B}}$. ■

Notation 13.1.2

For $J \in \mathfrak{M}_{n,n}(F)$, $B_{\mathcal{E}}^J(X,Y) = X^t J Y$ is a bilinear form on F^n with $[B_{\mathcal{E}}^J]_{\mathcal{E}} = J$.

Example 13.1.3

(a) $B_{\mathcal{E}}^I$ is the **ordinary inner product** X^tY of F^n (the dot product on $\mathbb{R}^n$; on $\mathbb{C}^n$ it is not Hermitian, but it is one of our objects of study). (b) The inner product $\sum j a_j b_j$ of Exercise 10.1.8 is $B_{\mathcal{E}}^J$ with $J = \operatorname{diag}(1,2,\dots,n)$. (c) The inner product of Exercise 10.1.9 is $B_{\mathcal{E}}^J$ with $J = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$.

Notation 13.1.4

For a basis $\mathfrak{B}$ of V and $J \in \mathfrak{M}_{n,n}(F)$, $B_{\mathfrak{B}}^J(v,w) = [v]_{\mathfrak{B}}^t J [w]_{\mathfrak{B}}$ is a bilinear form on V with $[B_{\mathfrak{B}}^J]_{\mathfrak{B}} = J$.

So $B \leftrightarrow [B]_{\mathfrak{B}}$ is a bijection between bilinear forms on V and $\mathfrak{M}_{n,n}(F)$ (for fixed $\mathfrak{B}$).

Exercise 13.1.5

$\mathfrak{Bil}(V) = \{B : V\times V \to F \text{ bilinear}\}$, with $(B+C)(v,w) = B(v,w)+C(v,w)$ and $(aB)(v,w) = aB(v,w)$, is an F-vector space.

Solution. $B + C$ and aB are bilinear; the axioms hold pointwise. □

Theorem 13.1.6 (Fundamental theorem for bilinear forms)

For a fixed basis $\mathfrak{B}$ of n-dimensional V, $\Omega_{\mathfrak{B}} : \mathfrak{Bil}(V) \to \mathfrak{M}_{n,n}(F)$, $\Omega_{\mathfrak{B}}(B) = [B]_{\mathfrak{B}}$, is a vector space isomorphism.

Proof. Injective by Observation 13.1.1, surjective by Notation 13.1.4, and linear since $[B+C]_{\mathfrak{B}} = [B]_{\mathfrak{B}} + [C]_{\mathfrak{B}}$ and $[aB]_{\mathfrak{B}} = a[B]_{\mathfrak{B}}$ (entrywise). ■

So, given $\mathfrak{B}$, a bilinear form "is" a matrix.

Remark 13.1.7

With the coordinate isomorphism $\alpha_{\mathfrak{B}} : V \to F^n$, the square

$$\begin{array}{ccc} V \times V & \xrightarrow{B} & F \\ {\scriptstyle \alpha_{\mathfrak{B}} \times \alpha_{\mathfrak{B}}}\downarrow & & \downarrow{\scriptstyle I} \\ F^n \times F^n & \xrightarrow[B^J_{\mathcal{E}}]{} & F \end{array}$$

($J = [B]_{\mathfrak{B}}$) commutes, which is Observation 13.1.1. So (V, B) and $(F^n, B^J_{\mathcal{E}})$ are essentially the same.

Observation 13.1.8

If J is invertible, then $B^J_{\mathcal{E}}(AX, Y) = B^J_{\mathcal{E}}(X, (J^{-1}A^tJ)Y)$ for $A \in \mathfrak{M}_{n,n}(F)$, $X, Y \in F^n$.

Proof. $(AX)^tJY = X^tA^tJY$ and $X^tJ(J^{-1}A^tJ)Y = X^tA^tJY$. ■

(This generalizes Observation 10.7.1; compare Question 10.7.2.)

Definition 13.1.9 (Symmetric, alternating)

A bilinear form B is (a) **symmetric** if $B(v, w) = B(w, v)$;
(b) **alternating** if $B(v, v) = 0$ for all v.

Observation 13.1.10

B is symmetric iff $[B]_{\mathfrak{B}}$ is a symmetric matrix.

Proof. If B is symmetric, $B(v_i, v_j) = B(v_j, v_i)$. Conversely, $(*)$ shows $B(v, w) = B(w, v)$ when $B(v_i, v_j) = B(v_j, v_i)$. ■

Exercise 13.1.11

B is alternating iff $B(v, w) = -B(w, v)$ for all v, w (when $1 + 1 \neq 0$ in F).

Solution. If alternating, $0 = B(v + w, v + w) = B(v, w) + B(w, v)$. Conversely, $B(v, v) = -B(v, v)$ gives $2B(v, v) = 0$. (In characteristic 2 only "alternating ⇒ skew" holds.) □

Exercise 13.1.12

Define an **alternating matrix** ($J^t = -J$ with zero diagonal) and show: B is alternating iff $[B]_{\mathfrak{B}}$ is alternating. (For $1 + 1 \neq 0$, alternating = skew-symmetric.)

Solution. If B is alternating, $B(v_i, v_i) = 0$ and $B(v_i, v_j) = -B(v_j, v_i)$. Conversely, by $(*)$, $B(v, v) = \sum_i a_i^2 J_{ii} + \sum_{i<j} a_i a_j (J_{ij} + J_{ji}) = 0$. □

Exercise 13.1.13

Let $\mathfrak{Sym}(V)$ and $\mathfrak{Alt}(V)$ be the symmetric and the alternating bilinear forms on V ($1+1 \neq 0$). (a) They are subspaces of $\mathfrak{Bil}(V)$. (b) $\dim \mathfrak{Bil}(V) = n^2$, $\dim \mathfrak{Sym}(V) = \frac{n(n+1)}{2}$, $\dim \mathfrak{Alt}(V) = \frac{n(n-1)}{2}$. (c) $\mathfrak{Bil}(V) = \mathfrak{Sym}(V) \oplus \mathfrak{Alt}(V)$.

Solution. (a) The defining conditions are linear.

(b) Via $\Omega_{\mathfrak{B}}$: symmetric matrices are determined by the entries on and above the diagonal, alternating ones by those strictly above.

(c) $B = \frac{1}{2}(B+B^t)+\frac{1}{2}(B-B^t)$ with $B^t(v,w) = B(w,v)$, and a form both symmetric and alternating is 0. □

Question 13.1.14

For $F = \mathbb{R}$, how can we tell from $[B]_{\mathfrak{B}}$ whether a symmetric bilinear form is an inner product (positive definite)? (Answered in §15.3.)

13.2 Quadratic forms

Motivation. A symmetric bilinear form is the same thing as a quadratic form $Q(v) = \frac{1}{2}B(v,v)$, i.e. a homogeneous quadratic polynomial in the coordinates. Chapters 9 and 10 were really about quadratic expressions.

Throughout, $1+1 \neq 0$ in F.

Example 13.2.1

For $J = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$, $B^J_{\mathcal{E}}((x,y)^t,(x,y)^t) = 2(x^2 - xy + y^2)$, and $Q(x,y) = x^2 - xy + y^2$ is the familiar quadratic form.

Definition 13.2.2 (Quadratic form)

$Q : V \to F$ is a **quadratic form** if (Q1) $B_Q(v,w) = Q(v+w) - Q(v) - Q(w)$ is a symmetric bilinear form, and (Q2) $Q(cv) = c^2 Q(v)$.

Exercise 13.2.3

For a symmetric bilinear form B, $Q(v) = \frac{1}{2}B(v,v)$ is a quadratic form with $B_Q = B$.

Solution. $Q(v+w) - Q(v) - Q(w) = \frac{1}{2}(B(v,w) + B(w,v)) = B(v,w)$, and $Q(cv) = c^2 Q(v)$. □

Observation 13.2.4

$Q \mapsto B_Q$ is a bijection between quadratic forms and symmetric bilinear forms on V.

Proof. $B_Q(v,v) = Q(2v) - 2Q(v) = 4Q(v) - 2Q(v) = 2Q(v)$, so Q is recovered from B_Q: injective. Surjective by Exercise 13.2.3. ■

Exercise 13.2.5

The quadratic forms on V form a vector space, and $Q \mapsto B_Q$ is a vector space isomorphism.

Solution. Pointwise operations preserve (Q1), (Q2), and $B_{aQ+Q'} = aB_Q + B_{Q'}$. □

Why "quadratic". With $[B_Q]_{\mathfrak{B}} = J = (a_{ij})$ (symmetric) and $[v]_{\mathfrak{B}} = (x_1, \dots, x_n)^t$,

$$2Q(v) = B_Q(v,v) = \sum_{i,j} a_{ij} x_i x_j = \sum_i a_{ii} x_i^2 + 2 \sum_{i<j} a_{ij} x_i x_j,$$

a homogeneous quadratic polynomial in the coordinates. Conversely every homogeneous quadratic $2Q = \sum_i a_{ii} x_i^2 + 2\sum_{i<j} a_{ij} x_i x_j$ gives, with $a_{ji} = a_{ij}$, $B_Q(v,w) = \sum_{i,j} x_i a_{ij} y_j = [v]^t J [w]$. So symmetric bilinear forms, quadratic forms and homogeneous quadratic polynomials are the same thing.

Definition 13.2.6 (Quadratic space)

A vector space with a symmetric bilinear form B (hence a quadratic form Q) is a **quadratic space** $(V, B) = (V, B, Q)$.

13.3 Orthogonal groups and symplectic groups

Motivation. The orthogonal group of a quadratic space consists of the invertible operators preserving B; in a basis it is $\{A : A^t J A = J\}$, which settles Remark 10.6.4. Replacing symmetric by alternating forms gives the symplectic groups.

Definition 13.3.1 (Orthogonal group of a quadratic space)

$\mathbf{O}(V,B) = \{L \in \mathrm{GL}(V) : B(Lv, Lw) = B(v,w)\ \forall v, w\}$, the **orthogonal group**; its elements are **orthogonal operators**. Studying (V,B) and $\mathbf{O}(V,B)$ is **orthogonal geometry**.

(We consider only elements of $\mathrm{GL}(V)$ from the start; injectivity is not even needed, but we skip that.)

Exercise 13.3.2

(a) $\mathbf{O}(V,B) \leq \mathrm{GL}(V)$.
(b) $\mathbf{SO}(V,B) = \{L \in \mathbf{O}(V,B) : \det L = 1\} \trianglelefteq \mathbf{O}(V,B)$.

Solution. (a) I preserves B; if L, M do, so do LM and L^{-1} ($B(L^{-1}v, L^{-1}w) = B(LL^{-1}v, LL^{-1}w)$).
(b) It is the kernel of $\det : \mathbf{O}(V,B) \to F^\times$. □

Exercise 13.3.3

(a) $\mathbf{O}(V,B) = \{L \in \mathrm{GL}(V) : Q(Lv) = Q(v)\ \forall v\}$.
(b) If $\mathfrak{B} = \{v_i\}$ is a basis, $\mathbf{O}(V,B) = \{L \in \mathrm{GL}(V) : B(Lv_i, Lv_j) = B(v_i, v_j)\ \forall i, j\}$.

Solution. (a) As in Observation 9.3.4: $B(v,w) = Q(v+w) - Q(v) - Q(w)$ is determined by Q.
(b) By bilinearity $(*)$. □

Exercise 13.3.4

Can $\mathbf{O}(V,B) = \mathrm{GL}(V)$?

Solution. Yes, iff $B = 0$ (or $V = 0$). If $B \neq 0$, choose v with $Q(v) \neq 0$ (Lemma: possible since $1+1 \neq 0$, cf. Exercise 13.2.3); then $L = 2I$ satisfies $Q(Lv) = 4Q(v) \neq Q(v)$ (as long as $3 \neq 0$; in characteristic 3 use $L = cI$ with $c^2 \neq 1$ if available, e.g. fields with more than 3 elements). For $B = 0$ every L preserves B. □

Proposition 13.3.5

Let $\mathfrak{B}$ be a basis of n-dimensional (V,B) and $J = [B]_{\mathfrak{B}}$. Then

$$\{[L]_{\mathfrak{B}}^{\mathfrak{B}} : L \in \mathbf{O}(V,B)\} = \{A \in \mathrm{GL}_n(F) : A^t J A = J\}.$$

Proof. By the fundamental theorem and Observation 13.1.1, $B(Lv,Lw) = B(v,w)$ for all v,w means $[Lv]_{\mathfrak{B}}^t J [Lw]_{\mathfrak{B}} = [v]_{\mathfrak{B}}^t J [w]_{\mathfrak{B}}$. Since $[Lv]_{\mathfrak{B}} = [L]_{\mathfrak{B}}^{\mathfrak{B}}[v]_{\mathfrak{B}}$, this is $[v]^t([L]^t J [L])[w] = [v]^t J [w]$ for all coordinate vectors, i.e. $[L]^t J [L] = J$ by Exercise 9.3.7. The argument is reversible. (Symmetry of J is not used.) ■

This settles Remark 10.6.4 ($J = I$ for an orthonormal basis).

Definition 13.3.6 (Matrix orthogonal groups)

For a symmetric $J \in \mathfrak{M}_{n,n}(F)$, $\mathbf{O}_n^J(F) = \{A \in \mathrm{GL}_n(F) : A^t J A = J\}$, the **orthogonal group** of J. (Thus $\mathbf{O}(n) = \mathbf{O}_n^I(\mathbb{R})$.)

$\mathbf{O}_n^J(F) \leq \mathrm{GL}_n(F)$, $\mathbf{SO}_n^J(F) \trianglelefteq \mathbf{O}_n^J(F)$, and with $J = [B]_{\mathfrak{B}}$, $\Psi_{\mathfrak{B}}^{\mathfrak{B}}$ restricts to $\mathbf{O}(V,B) \approx \mathbf{O}_n^J(F)$ (compare §11.9).

Exercise 13.3.7

If J is invertible symmetric and $A \in \mathbf{O}_n^J(F)$, then $\det A = \pm 1$.

Solution. $\det(A)^2 \det J = \det J \neq 0$. □

Definition 13.3.8 (Symplectic group)

For an alternating form B on V, $\mathbf{Sp}(V,B) = \{L \in \mathrm{GL}(V) : B(Lv,Lw) = B(v,w)\ \forall v,w\}$ is the **symplectic group**; this is **symplectic geometry**.

Proposition 13.3.9

If B is alternating, $\mathfrak{B}$ a basis and $J = [B]_{\mathfrak{B}}$ (skew-symmetric), then $\{[L]_{\mathfrak{B}}^{\mathfrak{B}} : L \in \mathbf{Sp}(V,B)\} = \{A \in \mathrm{GL}_n(F) : A^t J A = J\}$.

Proof. Word for word the proof of Proposition 13.3.5, which did not use symmetry of J. ■

Definition 13.3.10 (Matrix symplectic groups)

For skew-symmetric J, $\mathbf{Sp}_n^J(F) = \{A \in \mathrm{GL}_n(F) : A^t J A = J\}$.

Exercise 13.3.11

For $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $\mathbf{Sp}_2^J(F) = \mathrm{SL}_2(F)$.

Solution. For any 2×2 A, $A^t J A = (\det A)J$ (direct computation: $A^t J A$ is alternating, hence a multiple of J, and the $(1,2)$ entry is $ad - bc$). So $A^t J A = J$ iff $\det A = 1$. □

In general, for invertible skew-symmetric J (n even), $\mathbf{Sp}_n^J(F) \leq \mathrm{SL}_n(F)$; the proof is postponed (so there is no "**SSp**").

Question 13.3.12

For invertible symmetric J, is there always $A \in \mathbf{O}_n^J(F)$ with $\det A = -1$? How should "reflections" of a quadratic space be defined? (See §13.5.)

13.4 $\mathbf{O}(1,1)$ and $\mathbf{O}(3,1)$

Motivation. For the form $x^2 - y^2$ on $\mathbb{R}^2$, "rotations" are hyperbolic: cosh and sinh replace cos and sin, and the orthogonal group splits into four copies of a line instead of two circles. Adding three space coordinates gives the Lorentz group of special relativity.

For $J = \mathrm{diag}(1,-1)$, $B_{\mathcal{E}}^J((a,b)^t,(c,d)^t) = ac - bd$ (quadratic form $2Q = x^2 - y^2$), and

$$\mathbf{O}(1,1) = \mathbf{O}_2^J(\mathbb{R}) = \{A \in \mathrm{GL}_2(\mathbb{R}) : A^t J A = J\}.$$

$A = \left(\begin{smallmatrix} x & z \\ y & w \end{smallmatrix}\right) \in \mathbf{O}(1,1)$ iff $x^2 - y^2 = 1 = w^2 - z^2$ and $xz - yw = 0$: the columns lie on the hyperbolas $x^2 - y^2 = 1$, $w^2 - z^2 = 1$. Recall $\cosh t = \frac{e^t + e^{-t}}{2}$, $\sinh t = \frac{e^t - e^{-t}}{2}$.

Exercise 13.4.1

For $t, s \in \mathbb{R}$: (a) $\cosh^2 t - \sinh^2 t = 1$;
(b) $\cosh(-t) = \cosh t$, $\sinh(-t) = -\sinh t$;
(c) $\cosh(t+s) = \cosh t \cosh s + \sinh t \sinh s$;
(d) $\sinh(t+s) = \cosh t \sinh s + \sinh t \cosh s$.

Solution. Expand the exponentials; e.g. (a): $\frac{(e^t+e^{-t})^2 - (e^t - e^{-t})^2}{4} = 1$. □

Since $\cosh t \geq 1$, $t \mapsto (\cosh t, \sinh t)^t$ parametrizes the right branch of $x^2 - y^2 = 1$, and $(-\cosh t, \sinh t)^t$ the left one. If $x, w > 0$, write $(x,y) = (\cosh t, \sinh t)$, $(z,w) = (\sinh s, \cosh s)$; then $0 = xz - yw = \sinh(s-t)$, so $s = t$ and

$$A = \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix}.$$

Similarly: if $x > 0$, $w < 0$: $A = \left(\begin{smallmatrix} \cosh u & \sinh u \\ -\sinh u & -\cosh u \end{smallmatrix}\right)$; if $x < 0$, $w > 0$: $A = \left(\begin{smallmatrix} -\cosh t & -\sinh t \\ \sinh t & \cosh t \end{smallmatrix}\right)$; if $x, w < 0$: $A = \left(\begin{smallmatrix} -\cosh u & -\sinh u \\ -\sinh u & -\cosh u \end{smallmatrix}\right)$.

Exercise 13.4.2

With $\mathbf{SO}(1,1) = \mathrm{SL}_2(\mathbb{R}) \cap \mathbf{O}(1,1)$: $\mathbf{O}(1,1)/\mathbf{SO}(1,1) \approx \mu_2$, $[\mathbf{O}(1,1) : \mathbf{SO}(1,1)] = 2$, and $\mathbf{O}(1,1) = \mathbf{SO}(1,1) \amalg \mathrm{diag}(-1,1)\mathbf{SO}(1,1)$.

Solution. $\det : \mathbf{O}(1,1) \to \mu_2$ is onto ($\det \operatorname{diag}(-1,1) = -1$) with kernel $\mathbf{SO}(1,1)$; apply the First Isomorphism Theorem. □

Exercise 13.4.3

Let $\mathbf{SO}^\circ(1,1) = \left\{ \left(\begin{smallmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{smallmatrix}\right) \right\} = \left\{ \left(\begin{smallmatrix} x & y \\ y & x \end{smallmatrix}\right) : x \geq 1,\ x^2 - y^2 = 1 \right\}$. Then $\mathbf{SO}^\circ(1,1) < \mathbf{SO}(1,1)$, $\mathbf{SO}(1,1) = \mathbf{SO}^\circ(1,1) \amalg (-I \cdot \mathbf{SO}^\circ(1,1))$, $[\mathbf{SO}(1,1) : \mathbf{SO}^\circ(1,1)] = 2$, and $\mathbf{SO}^\circ(1,1) \trianglelefteq \mathbf{SO}(1,1)$.

Solution. The product of two such matrices is the matrix for $t+s$ (Exercise 13.4.1), and the inverse is the one for $-t$. From the four cases, $\det A = 1$ exactly in the cases $x, w > 0$ (which is $\mathbf{SO}^\circ$) and $x, w < 0$ (which is $-\mathbf{SO}^\circ$). Index 2 gives normality (Observation 12.2.9). □

Hence $[\mathbf{O}(1,1) : \mathbf{SO}^\circ] = 4$ (Observation 12.1.12) and, writing $\mathbf{SO}^\circ = \mathbf{SO}^\circ(1,1)$,

$$\mathbf{O}(1,1) = \mathbf{SO}^\circ \amalg (-I \cdot \mathbf{SO}^\circ) \amalg (\operatorname{diag}(-1,1)\mathbf{SO}^\circ) \amalg (\operatorname{diag}(1,-1)\mathbf{SO}^\circ),$$

matching the four cases.

Exercise 13.4.4

Is $\mathbf{SO}^\circ(1,1)$ normal in $\mathbf{O}(1,1)$?

Solution. Yes: $S = \operatorname{diag}(1,-1)$ gives $S\left(\begin{smallmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{smallmatrix}\right)S^{-1} = \left(\begin{smallmatrix} \cosh t & -\sinh t \\ -\sinh t & \cosh t \end{smallmatrix}\right) \in \mathbf{SO}^\circ$, and $\mathbf{O}(1,1)$ is generated by $\mathbf{SO}(1,1)$ (in which $\mathbf{SO}^\circ$ is normal) and S. □

Exercise 13.4.5

$\varphi : (\mathbb{R}, +) \to \mathbf{SO}^\circ(1,1)$, $\varphi(t) = \left(\begin{smallmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{smallmatrix}\right)$, is a group isomorphism.

Solution. A homomorphism by Exercise 13.4.1(c), (d); onto by the parametrization; injective since $\sinh t = 0$ only for $t = 0$. □

So $\mathbf{SO}^\circ(1,1)$ looks like $\mathbb{R}$, and $\mathbf{O}(1,1)$ is a disjoint union of four "real lines", whereas $\mathbf{O}(2)$ is a union of two circles.

Exercise 13.4.6

$\mathbf{O}(2) \not\approx \mathbf{O}(1,1)$.

Solution. [Supplement] Elements of finite order: $\mathbf{O}(2)$ contains $R_{2\pi/3}$, of order 3. In $\mathbf{O}(1,1)$ an element of finite order lies in $\mathbf{SO}^\circ(1,1)$ times one of $\pm I, \operatorname{diag}(\pm 1, \mp 1)$. $\mathbf{SO}^\circ \approx (\mathbb{R}, +)$ is torsion-free, so a power A^2 of $A \in \mathbf{O}(1,1)$ lies in $\mathbf{SO}^\circ$ (index 4, quotient $\mu_2 \times \mu_2$), and $A^{2k} = 1$ forces $A^2 = 1$. Hence all elements of finite order in $\mathbf{O}(1,1)$ have order 1 or 2, and isomorphisms preserve orders. □

Definition 13.4.7 (Lorentz group)

For $J = \operatorname{diag}(1,1,1,-1)$, $\mathbf{O}(3,1) = \mathbf{O}_4^J(\mathbb{R})$ is the **Lorentz group**, the symmetry group of space-time $\mathbb{R}^4$.

Exercise 13.4.8

$\mathbf{O}(3)$ and $\mathbf{O}(1,1)$ are isomorphic to subgroups of $\mathbf{O}(3,1)$.

Solution. $A \mapsto \operatorname{diag}(A, 1)$ embeds $\mathbf{O}(3)$; $\left(\begin{smallmatrix} x & z \\ y & w \end{smallmatrix}\right) \mapsto \operatorname{diag}(1, 1, \cdot)$ with the 2×2 block in positions $3, 4$ embeds $\mathbf{O}(1,1)$. Both preserve $A^t J A = J$ blockwise. □

13.5 Non-degenerate bilinear forms

Motivation. Without positive definiteness a vector can be perpendicular to itself, and $W \oplus W^\perp = V$ may fail. What survives is $\dim W^\perp = \dim V - \dim W$ for non-degenerate forms, together with orthogonal bases of quadratic spaces and reflections.

Throughout, V is n-dimensional with a bilinear form B, and $1 + 1 \neq 0$.

Definition 13.5.1 (Orthogonality)

$v \perp w$ if $B(v, w) = 0$; $S \perp T$ if $B(v, w) = 0$ for all $v \in S$, $w \in T$.

Remark 13.5.2

If B is symmetric or alternating, $v \perp w \iff w \perp v$. Conversely one can show that $\perp$ is symmetric only for such B; this is why we study orthogonal and symplectic geometry.

Definition 13.5.3 (Orthogonal basis)

A basis $\mathfrak{B} = \{v_i\}$ of mutually perpendicular vectors is an **orthogonal basis**; if moreover $B(v_i, v_i) = 1$ for all i, an **orthonormal basis**.

Exercise 13.5.4

(a) $\mathfrak{B}$ is orthogonal iff $[B]_\mathfrak{B}$ is diagonal.
(b) $\mathfrak{B}$ is orthonormal iff $[B]_\mathfrak{B} = I$.

Solution. The entries of $[B]_\mathfrak{B}$ are $B(v_i, v_j)$. □

Now $B(v, v) = 0$ is possible for $v \neq 0$, so orthogonal sets need not be independent, orthogonal bases cannot always be normalized, and Gram–Schmidt is unavailable.

Notation 13.5.5

For $S \subseteq V$, $S^\perp = \{v \in V : B(w, v) = 0 \text{ for all } w \in S\}$.

Exercise 13.5.6

(a) $S^\perp = \langle S \rangle^\perp \leq V$.
(b) If B is symmetric or alternating, $W \leq (W^\perp)^\perp$.

Solution. (a) Linearity in the second variable, and Exercise 10.2.16.

(b) For $w \in W$, $v \in W^\perp$: $B(v, w) = \pm B(w, v) = 0$. □

If $B = 0$, then $W^\perp = V$, so the perp theorem fails in general.

Exercise 13.5.7

If $B(w, w) \neq 0$, then $\dim \langle w \rangle^\perp = \dim V - 1$.

Solution. $v \mapsto B(w, v)$ is a nonzero linear functional (it is $\neq 0$ at w); its kernel $\langle w \rangle^\perp$ has dimension $n - 1$. □

Question 13.5.8

When is $\dim W^\perp = \dim V - \dim W$ for $W \leq V$?

Definition 13.5.9 (Non-degenerate)

B is **non-degenerate** if $V^\perp = 0$.

(Exercises 9.1.3, 10.2.1 said inner products are non-degenerate; the ordinary inner product $B^I_{\mathcal{E}}$ is non-degenerate.)

Proposition 13.5.10

For a basis $\mathfrak{B}$ of V: B is non-degenerate iff $\det[B]_{\mathfrak{B}} \neq 0$.

First proof (elementary). With $J = [B]_{\mathfrak{B}}$: B is degenerate iff some $0 \neq Y$ has $X^tJY = 0$ for all $X \in F^n$ (Observation 13.1.1), iff (taking $X = e_i$) some $0 \neq Y$ has $JY = 0$, iff $\det J = 0$. ■

(A second proof via the dual space is given after Observation 13.8.3.)

Theorem 13.5.11 (Perp theorem, non-degenerate case)

If B is non-degenerate and $W \leq V$, then $\dim W^\perp = \dim V - \dim W$. (This does *not* say $V = W \oplus W^\perp$.)

Proof. Extend a basis $\{v_1, \ldots, v_m\}$ of W to a basis $\mathfrak{B}$ of V, $J = [B]_{\mathfrak{B}}$. Then $v \in W^\perp$ iff $0 = [v_i]^tJ[v] = e_i^tJ[v] = [J]_i[v]$ for $i \leq m$. With K the $m \times n$ matrix of the first m rows of J, $v \in W^\perp$ iff $[v]_{\mathfrak{B}} \in \ker L_K$, so $\dim W^\perp = \dim \ker L_K$. J is invertible, so its rows are independent and $\operatorname{rk} K = m$. By the rank theorem $\dim \operatorname{im} L_K = m$, and by the dimension theorem $\dim \ker L_K = n - m$. ■

Proposition 13.5.12 (The three pillars)

Of (1) the dimension theorem (for L_A), (2) the rank theorem, (3) the perp theorem (non-degenerate case), any two imply the third.

Proof. As in Proposition 10.3.11, using the ordinary inner product $B^I_{\mathcal{E}}$ on F^n (non-degenerate): the row space W of A has $W^\perp = \ker L_A$, and the three statements are the three linear relations among n, $\dim \ker L_A$, $\dim W$ and $\dim \operatorname{im} L_A$. ■

These proofs do not depend on F (the rank theorem's proof in §10.3 for $F = \mathbb{C}$ needed a trick because the Hermitian dot product was used). A proof via the dual space follows in §13.8.

For the rest of this section (V, B) is a quadratic space (B symmetric).

Exercise 13.5.13

If (V, B) is a non-degenerate quadratic space and $U, W \leq V$: (a) $(W^\perp)^\perp = W$;
(b) $(U + W)^\perp = U^\perp \cap W^\perp$;
(c) $U^\perp + W^\perp = (U \cap W)^\perp$.

Solution. As in Exercise 10.3.6, using Theorem 13.5.11 and Exercise 13.5.6(b). □

Lemma 13.5.14

If B is symmetric, $w \in V$ and $B(w,w) \neq 0$, then $V = \langle w \rangle \oplus \langle w \rangle^{\perp}$.

Proof. For $v \in V$, $v = \frac{B(v,w)}{B(w,w)}w + \left(v - \frac{B(v,w)}{B(w,w)}w\right)$ with the second term in $\langle w \rangle^{\perp}$. And $cw \in \langle w \rangle^{\perp}$ means $cB(w,w) = 0$, so $c = 0$. ■

Observation 13.5.15

If B is symmetric and $B(v,v) = 0$ for all v, then $B = 0$.

Proof. $2B(v,w) = B(v+w, v+w) - B(v,v) - B(w,w) = 0$. ■

Theorem 13.5.16

Every quadratic space (V,B) ($V \neq 0$) has an orthogonal basis (B need not be non-degenerate).

Proof. If $B = 0$, every basis is orthogonal. Otherwise there is w with $B(w,w) \neq 0$ (Observation 13.5.15), and $V = \langle w \rangle \oplus \langle w \rangle^{\perp}$ (Lemma 13.5.14). By induction on dimension, $\langle w \rangle^{\perp}$ (with B restricted) has an orthogonal basis; add w. ■

Definition 13.5.17 (Reflection)

For symmetric B and w with $B(w,w) \neq 0$, $S_w(v) = v - \frac{2B(v,w)}{B(w,w)}w$ is the **reflection** (**symmetry**) with respect to w.

Exercise 13.5.18

For symmetric B and $B(w,w) \neq 0$: (a) S_w is linear and $S_w \in \mathbf{O}(V,B)$;
(b) $S_w = S_{cw}$ for $c \neq 0$;
(c) $S_w(w) = -w$;
(d) $S_w(v) = v$ for $v \in \langle w \rangle^{\perp}$.

Solution. (a) Linearity is clear. $B(S_w v, S_w u) = B(v,u) - \frac{2B(v,w)B(w,u)}{B(w,w)} - \frac{2B(u,w)B(v,w)}{B(w,w)} + \frac{4B(v,w)B(u,w)}{B(w,w)^2}B(w,w) = B(v,u)$, using symmetry ($B(w,u) = B(u,w)$). (b)–(d) as in Exercise 9.4.4. □

By Lemma 13.5.14, S_w is the unique linear operator satisfying (c), (d). With an orthogonal basis $\mathfrak{C} = \{v_2, \ldots, v_n\}$ of $\langle w \rangle^{\perp}$, $\mathfrak{B} = \{w, v_2, \ldots, v_n\}$ is an orthogonal basis, $[S_w]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(-1,1,\ldots,1) \in \mathbf{O}_n^J(F)$ (Proposition 13.3.5, $J = [B]_{\mathfrak{B}}$ diagonal), $\det S_w = -1$ and $S_w^2 = I$.

Exercise 13.5.19

For a quadratic space (V,B): (a) some $L \in \mathbf{O}(V,B)$ has $\det L = -1$;
(b) if B is non-degenerate, $\mathbf{O}(V,B)/\mathbf{SO}(V,B) \approx \mu_2$.

Solution. (a) If $B \neq 0$, a reflection S_w. If $B = 0$, $\mathbf{O}(V,B) = \mathrm{GL}(V)$ contains $\operatorname{diag}(-1,1,\ldots)$.

(b) $\det : \mathbf{O}(V,B) \to F^{\times}$ takes values ± 1 (Exercise 13.3.7) and is onto μ_2 by (a); apply the First Isomorphism Theorem. □

Exercise 13.5.20

(a) $-1+2\cosh^2\frac{t}{2}=\cosh t=1+2\sinh^2\frac{t}{2}$ and $2\cosh\frac{t}{2}\sinh\frac{t}{2}=\sinh t$. (b) For $J=\mathrm{diag}(1,-1)$ and $Y=(\cosh\frac{t}{2},-\sinh\frac{t}{2})^t$, $B^J_{\mathcal{E}}(Y,Y)=1$ and $[S_Y]_{\mathcal{E}}=\left(\begin{smallmatrix}-\cosh t & -\sinh t\\ \sinh t & \cosh t\end{smallmatrix}\right)$, so this matrix is a reflection of $\mathbf{O}(1,1)$.

Solution. (a) Exercise 13.4.1(c),

(d) with $s=t$, replaced by $t/2$. (b) $B(Y,Y)=\cosh^2\frac{t}{2}-\sinh^2\frac{t}{2}=1$. $S_Ye_1=e_1-2\cosh\frac{t}{2}Y=(1-2\cosh^2\frac{t}{2},\ 2\cosh\frac{t}{2}\sinh\frac{t}{2})^t=(-\cosh t,\sinh t)^t$, and $S_Ye_2=e_2-2B(e_2,Y)Y=e_2-2\sinh\frac{t}{2}Y=(-2\sinh\frac{t}{2}\cosh\frac{t}{2},\ 1+2\sinh^2\frac{t}{2})^t=(-\sinh t,\cosh t)^t$. □

The orthogonal-geometry story continues with "every orthogonal operator is a product of finitely many reflections" (compare Exercise 9.5.15).

13.6 Dual spaces and dual maps

Motivation. The dual space V^* of linear functionals, its dual basis, the natural isomorphism $V\cong V^{**}$ and the dual map $L^*(g)=g\circ L$ are the tools for a basis-free treatment of forms. The matrix of a dual map is the transpose: transposition is born here.

$V^*=\mathfrak{L}(V,F)$ is the **dual space**, its elements **linear functionals**; $V^{**}=(V^*)^*$ is the **double dual**. For a basis $\mathfrak{B}=\{v_1,\dots,v_n\}$, $v_i^*(v_j)=\delta_{ij}$ defines the **dual basis** $\mathfrak{B}^*=\{v_1^*,\dots,v_n^*\}$ of V^* (§5.1), so $\dim V=\dim V^*=\dim V^{**}$.

Proposition 13.6.1

$\psi^V:V\to V^{**}$, $\psi^V(v)(f)=f(v)$, is a vector space isomorphism, independent of any choice of basis (a **natural isomorphism**); we use it to identify V and V^{**}.

Proof. $\psi^V(v)$ is linear in f, so $\psi^V(v)\in V^{**}$; ψ^V is linear in v. If $\psi^V(v)=0$, then $f(v)=0$ for all $f\in V^*$, so $v=0$ (Observation 13.6.2). As $\dim V=\dim V^{**}$, the dimension theorem gives surjectivity. ■

Observation 13.6.2

If $f(v)=0$ for all $f\in V^*$, then $v=0$.

Proof. With $v=\sum a_iv_i$, $0=v_i^*(v)=a_i$ for all i. ■

Observation 13.6.3

For a basis $\{v_i\}$ of V, $\psi^V(v_i)=v_i^{**}$ (the dual basis of $\{v_i^*\}$).

Proof. $\psi^V(v_i)(v_j^*)=v_j^*(v_i)=\delta_{ij}=v_i^{**}(v_j^*)$; two linear maps agreeing on a basis are equal. ■

Notation 13.6.4

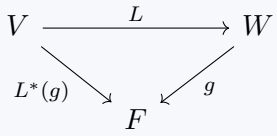
When there is a natural isomorphism, we identify the two spaces and abuse notation: $V=V^{**}$, $v_i=v_i^{**}$.

Remark 13.6.5

The isomorphisms of the first and second isomorphism theorems (Theorems 12.4.14, 12.4.21) are also natural.

Definition 13.6.6 (Dual map)

For linear $L : V \to W$, $L^* : W^* \to V^*$, $L^*(g) = g \circ L$, is the **dual map** of L.



Exercise 13.6.7

(a) For $g \in W^*$, $g \circ L \in V^*$.
(b) L^* is linear.

Solution. (a) A composite of linear maps.

(b) $(ag + h) \circ L = a(g \circ L) + h \circ L$. □

Observation 13.6.8

For bases $\mathfrak{B} = \{v_j\}$ of V, $\mathfrak{C} = \{w_i\}$ of W and $L \in \mathfrak{L}(V, W)$: $[L^*]_{\mathfrak{C}^*}^{\mathfrak{B}^*} = ([L]_{\mathfrak{B}}^{\mathfrak{C}})^t$.

Proof. Let $[L]_{\mathfrak{B}}^{\mathfrak{C}} = (a_{ij})$, i.e. $Lv_j = \sum_i a_{ij} w_i$. We must show $L^* w_i^* = \sum_j a_{ij} v_j^*$, i.e. $(L^* w_i^*)(v_\ell) = a_{i\ell}$. Indeed $(L^* w_i^*)(v_\ell) = w_i^*(Lv_\ell) = w_i^*\left(\sum_k a_{k\ell} w_k\right) = a_{i\ell}$. ■

So the transpose of a matrix corresponds to the dual of a linear map; this answers Question 5.4.22. (The meaning of $L^t \in \mathfrak{L}(W, V)$ itself is still open.)

Exercise 13.6.9

For $L \in \mathfrak{L}(V, W)$, $M \in \mathfrak{L}(W, U)$: (a) $(I_V)^* = I_{V^*}$;
(b) $(M \circ L)^* = L^* \circ M^*$;
(c) if L is invertible, so is L^*, with $(L^*)^{-1} = (L^{-1})^*$.

Solution. (a) $g \circ I = g$.

(b) $(ML)^*(g) = g \circ M \circ L = L^*(M^*(g))$.

(c) Apply (a), (b) to $LL^{-1} = I$ and $L^{-1}L = I$. □

Exercise 13.6.10

(a) $(L_1 + L_2)^* = L_1^* + L_2^*$;
(b) $(cL)^* = cL^*$;
(c) $* : \mathfrak{L}(V, W) \to \mathfrak{L}(W^*, V^*)$, $L \mapsto L^*$, is an isomorphism.

Solution. (a), (b): $g \circ (L_1 + L_2) = g \circ L_1 + g \circ L_2$, $g \circ (cL) = c(g \circ L)$.

(c) $*$ is linear by (a), (b), and via Observation 13.6.8 it corresponds to the transpose $\mathfrak{M}_{m,n}(F) \to \mathfrak{M}_{n,m}(F)$, which is bijective. □

Exercise 13.6.11

Translate Exercises 13.6.9 and 13.6.10 into matrix language.

Solution. $I^t = I$, $(AB)^t = B^t A^t$, $(A^{-1})^t = (A^t)^{-1}$, $(A+B)^t = A^t + B^t$, $(cA)^t = cA^t$, and $A \mapsto A^t$ is an isomorphism $\mathfrak{M}_{m,n}(F) \to \mathfrak{M}_{n,m}(F)$. □

Exercise 13.6.12

For $L \in \mathfrak{L}(V, W)$, the square

$$\begin{array}{ccc} V & \xrightarrow{L} & W \\ {\scriptstyle \psi^V}\downarrow & & \downarrow{\scriptstyle \psi^W} \\ V^{**} & \xrightarrow[L^{**}]{} & W^{**} \end{array}$$

commutes; so with $V = V^{**}$, $W = W^{**}$, $L = L^{**}$ (in matrices, $A = (A^t)^t$).

Solution. For $v \in V$, $g \in W^*$: $L^{**}(\psi^V v)(g) = \psi^V v(L^* g) = (g \circ L)(v) = g(Lv) = \psi^W(Lv)(g)$. □

13.7 Duality

Motivation. The natural pairing $\varepsilon(v, f) = f(v)$ between V and V^* behaves like a non-degenerate bilinear form. Annihilators satisfy $\dim W^{\text{perp}} = \dim V - \dim W$, and the dual statements follow by symmetry through $V = V^{**}$.

V comes with the bilinear form $\varepsilon : V \times V^* \to F$, $\varepsilon(v, f) = f(v)$ (a **bilinear pairing**); we write $v \perp f$ if $\varepsilon(v, f) = 0$.

Notation 13.7.1

For $S \subseteq V$, $T \subseteq V^*$: $S^{\text{perp}} = \{f \in V^* : \varepsilon(v, f) = 0 \ \forall v \in S\}$ and $T^{\text{perp}} = \{v \in V : \varepsilon(v, f) = 0 \ \forall f \in T\}$ (the **annihilators**).

Exercise 13.7.2

(a) $S^{\text{perp}} = \langle S \rangle^{\text{perp}} \leq V^*$.
(b) $T^{\text{perp}} = \langle T \rangle^{\text{perp}} \leq V$.

Solution. Linearity of ε in each variable. □

Theorem 13.7.3 (Duality)

For $W \leq V$, $Y \leq V^*$: (a) $\dim W^{\text{perp}} = \dim V - \dim W$;
(b) $\dim Y^{\text{perp}} = \dim V - \dim Y$.

Proof. (a) Extend a basis $\{v_1, \ldots, v_r\}$ of W to a basis $\{v_1, \ldots, v_n\}$ of V. Then $\{v_{r+1}^*, \ldots, v_n^*\}$ is a basis of W^{perp}: these vanish on W, and if $f = \sum a_i v_i^* \in W^{\text{perp}}$, then $a_i = f(v_i) = 0$ for $i \leq r$.

(b) (First proof.) Extend a basis $\{f_1, \ldots, f_s\}$ of Y to a basis $\{f_1, \ldots, f_n\}$ of V^* with dual basis $\{f_1^*, \ldots, f_n^*\}$ of V^{**}. Since ψ^V is an isomorphism, it suffices that $\psi^V(Y^{\text{perp}}) = \langle f_{s+1}^*, \ldots, f_n^* \rangle$. If $v \in Y^{\text{perp}}$ and $\psi^V(v) = \sum a_i f_i^*$, then $a_i = \psi^V(v)(f_i) = f_i(v) = 0$ for $i \leq s$. Conversely, $f_j^* = \psi^V(w_j)$ for a unique $w_j \in V$, and for $i \leq s < j$, $0 = f_j^*(f_i) = f_i(w_j)$, so $w_j \in Y^{\text{perp}}$. ■

(b) Second proof. V^* comes with $\delta : V^{**} \times V^* \to F$, $\delta(\alpha, f) = \alpha(f)$. Applying (a) to V^* and $Y \leq V^*$, the annihilator $Y^{\text{d-perp}} = \{\alpha \in V^{**} : \alpha(f) = 0 \ \forall f \in Y\}$ has dimension $\dim V^* - \dim Y = \dim V - \dim Y$. Under ψ^V, $\delta(\psi^V v, f) = f(v) = \varepsilon(v, f)$ (the square below commutes), so $\psi^V(Y^{\text{perp}}) = Y^{\text{d-perp}}$.

$$\begin{array}{ccc} V \times V^* & \xrightarrow{\ \varepsilon\ } & F \\ {\scriptstyle \psi^V \times I}\downarrow & & \downarrow{\scriptstyle I} \\ V^{**} \times V^* & \xrightarrow[\ \delta\ ]{} & F \end{array}$$

■

That a proper subspace $W \neq V$ has a nonzero functional vanishing on it is easy; its "dual statement" needs the theorem:

Corollary 13.7.4

If $Y \leq V^*$, $Y \neq V^*$, then some $0 \neq v \in V$ has $f(v) = 0$ for all $f \in Y$.

Proof. By duality, $\dim Y^{\text{perp}} = \dim V - \dim Y > 0$. ■

Exercise 13.7.5

For $W \leq V$: (a) $(W^{\text{perp}})^{\text{perp}} = W$;
(b) with $V = V^{**}$, $(W^{\text{perp}})^{\text{d-perp}} = W$.

Solution. $W \leq (W^{\text{perp}})^{\text{perp}}$ is clear, and the dimensions agree by Theorem 13.7.3 applied twice.
(b) is (a) transported by ψ^V. □

Exercise 13.7.6

For $U, W \leq V$: (a) $(U + W)^{\text{perp}} = U^{\text{perp}} \cap W^{\text{perp}}$;
(b) $(U \cap W)^{\text{perp}} = U^{\text{perp}} + W^{\text{perp}}$.

Solution. (a) is immediate.

(b) $\supseteq$ is clear; the dimensions agree by Theorem 13.7.3, (a) and Corollary 12.4.22: $\dim(U^{\text{perp}} + W^{\text{perp}}) = (n - u) + (n - w) - (n - \dim(U + W)) = n - \dim(U \cap W)$. □

Exercise 13.7.7

For $L \in \mathfrak{L}(V, W)$: (a) $\ker(L^*) = (\operatorname{im} L)^{\text{perp}}$;
(b) $\operatorname{im}(L^*) = (\ker L)^{\text{perp}}$.

Solution. (a) $L^* g = 0 \iff g(Lv) = 0 \ \forall v \iff g \in (\operatorname{im} L)^{\text{perp}}$. (b) $\operatorname{im} L^* \subseteq (\ker L)^{\text{perp}}$: $(g \circ L)(v) = 0$ for $v \in \ker L$. The dimensions agree: $\dim \operatorname{im} L^* = \dim W - \dim \ker L^* = \dim W - (\dim W - \operatorname{rk} L) = \operatorname{rk} L = \dim V - \dim \ker L$. □

(Compare Exercise 10.3.6 and Observation 10.7.9.)

13.8 B-identification

Motivation. A bilinear form B is the same as a linear map $\varphi_B : V \to V^*$, $v \mapsto B(\cdot, v)$, with matrix $[B]_{\mathfrak{B}}$. B is non-degenerate iff φ_B is an isomorphism; identifying $V = V^*$ through it turns B into

the canonical pairing ε, which proves the perp theorem in one line.

Definition 13.8.1 (φ_B)

For a bilinear form B on V, $\varphi_B^V : V \to V^*$, $\varphi_B^V(v)(w) = B(w,v)$, is a linear map (well defined because B is linear in the first variable, linear because it is linear in the second).

Theorem 13.8.2

The following are equivalent: (1) B is non-degenerate; (2) φ_B^V is a monomorphism; (3) φ_B^V is an isomorphism; (4) every element of V^* is $\varphi_B^V(v)$ for some v.

Proof. $\ker \varphi_B^V = \{v : B(w,v) = 0 \ \forall w\} = V^\perp$. So (1)⇔(2), and (2)⇔(3)⇔(4) by the dimension theorem ($\dim V = \dim V^*$). ■

The surjectivity of φ_B^V cannot be shown without the dimension theorem (and fails in infinite dimensions).

Observation 13.8.3

For a basis $\mathfrak{B}$ of V with dual basis $\mathfrak{B}^*$: $[\varphi_B^V]_{\mathfrak{B}}^{\mathfrak{B}^*} = [B]_{\mathfrak{B}}$.

Proof. With $[\varphi_B^V]_{\mathfrak{B}}^{\mathfrak{B}^*} = (b_{ij})$, $\varphi_B^V(v_j) = \sum_i b_{ij} v_i^*$, so $B(v_k, v_j) = \varphi_B^V(v_j)(v_k) = b_{kj}$. ■

So a bilinear form, an $n \times n$ matrix, and a linear map $V \to V^*$ are the same data once $\mathfrak{B}$ is fixed: dual spaces are unavoidable in the study of bilinear forms.

Second proof of Proposition 13.5.10. B is non-degenerate iff φ_B^V is an isomorphism, iff its matrix $[B]_{\mathfrak{B}}$ is invertible. ■

Exercise 13.8.4

Identifying $V = V^{**}$, the dual map $(\varphi_B^V)^* : V^{**} \to V^*$ becomes a map $V \to V^*$. Show that B is symmetric iff $(\varphi_B^V)^* = \varphi_B^V$.

Solution. $(\varphi_B^V)^*(\psi^V v)(w) = \psi^V(v)(\varphi_B^V w) = \varphi_B^V(w)(v) = B(v,w)$, while $\varphi_B^V(v)(w) = B(w,v)$. These agree for all v, w iff B is symmetric. (In matrices: $[B]^t = [B]$.) □

Notation 13.8.5 (B-identification)

If B is non-degenerate, φ_B^V is a natural isomorphism, and we identify $V = V^*$ through it, writing $v = \varphi_v = \varphi_B^V(v)$.

Remark 13.8.6

Transport B to V^*: $B^*(\varphi_v, \varphi_w) = B(w,v)$ is a non-degenerate bilinear form on V^*, giving $V^* = V^{**}$. For the identifications to be consistent we need $\varphi_{B^*}^{V^*} \circ \varphi_B^V = \psi^V$. Indeed $\varphi_{B^*}^{V^*}(\varphi_B^V(v))(\varphi_w) = B^*(\varphi_w, \varphi_v) = B(v,w) = \varphi_w(v) = \psi^V(v)(\varphi_w)$.

Exercise 13.8.7

For $f \in (\mathbb{R}^2)^*$, $f((x,y)^t) = x + y$, find the vector of $\mathbb{R}^2$ corresponding to f under $B^J_{\mathcal{E}}$-identification, for (a) $J = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$,
(b) $J = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$.

Solution. We need v with $X^t J v = x + y$, i.e. $Jv = (1,1)^t$. (a) $v = (1,-1)^t$.
(b) $v = J^{-1}(1,1)^t = \frac{1}{3}\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)(1,1)^t = (1,1)^t$. □

Exercise 13.8.8

If B is non-degenerate and $\mathfrak{B} = \{v_i\}$ is a basis, there is a basis $\mathfrak{C} = \{w_j\}$ with $B(v_i, w_j) = \delta_{ij}$, and $[I]^{\mathfrak{B}}_{\mathfrak{C}} = [B]^{-1}_{\mathfrak{B}}$.

Solution. Let $w_j = (\varphi^V_B)^{-1}(v_j^*)$; then $B(v_i, w_j) = v_j^*(v_i) = \delta_{ij}$ (the basis "dual" to $\mathfrak{B}$ under the identification). Write $w_j = \sum_k c_{kj} v_k$, $C = [I]^{\mathfrak{B}}_{\mathfrak{C}} = (c_{kj})$. Then $\delta_{ij} = \sum_k B(v_i, v_k) c_{kj} = ([B]_{\mathfrak{B}} C)_{ij}$, so $C = [B]^{-1}_{\mathfrak{B}}$. □

Exercise 13.8.9

If V has an orthonormal basis $\{v_i\}$ (so (V,B) is a non-degenerate quadratic space), then $\varphi^V_B(v_i) = v_i^*$.

Solution. $\varphi^V_B(v_i)(v_k) = B(v_k, v_i) = \delta_{ki} = v_i^*(v_k)$. (Compare Observation 13.6.3.) □

Proposition 13.8.10

If B is non-degenerate, the square

$$\begin{array}{ccc} V \times V & \xrightarrow{B} & F \\ {\scriptstyle I \times \varphi^V_B}\downarrow & & \downarrow{\scriptstyle I} \\ V \times V^* & \xrightarrow[\varepsilon]{} & F \end{array}$$

commutes: under B-identification, $B = \varepsilon$. In particular "$\perp$= perp".

Proof. $\varepsilon(v, \varphi^V_B w) = \varphi^V_B(w)(v) = B(v,w)$. ■

Theorem 13.8.11 (Perp theorem, non-degenerate case)

If B is non-degenerate and $W \leq V$, then under B-identification $W^\perp = W^{\text{perp}}$; in particular $\dim W^\perp = \dim V - \dim W$.

Proof. By Proposition 13.8.10, $v \in W^\perp$ iff $\varphi^V_B(v) \in W^{\text{perp}}$, i.e. $\varphi^V_B(W^\perp) = W^{\text{perp}}$; apply Theorem 13.7.3(a). ■

Exercise 13.8.13

Let (V,B) be a non-degenerate quadratic space and $U \leq W \leq V$. Define $\vartheta : W \to (U^\perp/W^\perp)^*$, $\vartheta(w)(\bar{v}) = B(v,w)$ $(w \in W,\ v \in U^\perp)$. (a) ϑ is a well-defined linear map with $\ker \vartheta = U$. (b) There is a natural isomorphism $W/U \approx (U^\perp/W^\perp)^*$; in particular $W \approx (V/W^\perp)^*$. (c) There is a natural isomorphism $W/(W \cap W^\perp) \approx (W/(W \cap W^\perp))^*$.

Solution. (a) Note $W^\perp \le U^\perp$. If $\bar{v} = \bar{v}'$, then $v - v' \in W^\perp$, so $B(v - v', w) = 0$: well defined. Linear in w. $\vartheta(w) = 0$ iff $w \perp U^\perp$ iff $w \in (U^\perp)^\perp = U$ (Exercise 13.5.13). (b) By (a) and the First Isomorphism Theorem, $W/U \approx \operatorname{im}\vartheta$. The dimensions agree: $\dim(U^\perp/W^\perp) = (n-u) - (n-w) = w - u$ (Theorem 13.5.11), so ϑ is onto. Take $U = 0$ for the special case. (c) The restriction of B to W induces $\bar{B}(\bar{w}, \bar{w}') = B(w, w')$ on $W/(W \cap W^\perp)$; it is well defined (the radical of $B|_W$ is $W \cap W^\perp$) and non-degenerate, so $\varphi_{\bar{B}}$ is a natural isomorphism (Theorem 13.8.2). □

13.9 Transpose operators

Motivation. Combining dual maps with B-identification defines the transpose L^t of a linear map between quadratic spaces, characterised by $C(Lv, w) = B(v, L^t w)$. Its matrix is $J_{\mathfrak{B}}^{-1}[L]^t J_{\mathfrak{C}}$, and orthogonal operators are exactly those with $L^t L = I$. This is the long-promised meaning of the transpose.

Throughout, (V, B) and (W, C) are non-degenerate quadratic spaces.

Definition 13.9.1 (Transpose map)

(a) For $L \in \mathfrak{L}(V, W)$, the **transpose** $L^t \in \mathfrak{L}(W, V)$ (with respect to B, C) is L^* under B- and C-identification, i.e. the unique linear map making

$$\begin{array}{ccc} W & \xrightarrow{L^t} & V \\ {\scriptstyle \varphi_C^W}\downarrow & & \downarrow{\scriptstyle \varphi_B^V} \\ W^* & \xrightarrow[L^*]{} & V^* \end{array}$$

commute: $L^t = (\varphi_B^V)^{-1} \circ L^* \circ \varphi_C^W$.
(b) For $L \in \mathfrak{L}(V, V)$, the **transpose operator** is $L^t = (\varphi_B^V)^{-1} \circ L^* \circ \varphi_B^V$.

(The dual map does not depend on B, C; the transpose does.)

Observation 13.9.2

If $L, M \in \mathfrak{L}(V, V)$ and $B(Lv, w) = B(Mv, w)$ for all v, w, then $L = M$ (likewise in the second variable).

Proof. $B(Lv - Mv, w) = 0$ for all w, so $Lv - Mv \in V^\perp = 0$ (by symmetry of B). ■

Proposition 13.9.3

L^t is the unique linear map with $C(Lv, w) = B(v, L^t w)$ for all $v \in V$, $w \in W$ (equivalently $C(w, Lv) = B(L^t w, v)$).

Proof. The definition says $L^*(\varphi_C^W(w)) = \varphi_B^V(L^t w)$. Evaluating at v: $C(Lv, w) = \varphi_w(Lv) = (L^*\varphi_w)(v) = \varphi_{L^t w}(v) = B(v, L^t w)$. Uniqueness follows from Observation 13.9.2 (non-degeneracy). ■

Proposition 13.9.4

For bases $\mathfrak{B}, \mathfrak{C}$ of V, W with $J_{\mathfrak{B}} = [B]_{\mathfrak{B}}$, $J_{\mathfrak{C}} = [C]_{\mathfrak{C}}$: (a) $[L^t]_{\mathfrak{C}}^{\mathfrak{B}} = (J_{\mathfrak{B}})^{-1}([L]_{\mathfrak{B}}^{\mathfrak{C}})^t J_{\mathfrak{C}}$; (b) for $L \in \mathfrak{L}(V, V)$ and $J = [B]_{\mathfrak{B}}$, $[L^t]_{\mathfrak{B}}^{\mathfrak{B}} = J^{-1}([L]_{\mathfrak{B}}^{\mathfrak{B}})^t J$.

Proof. From $L^t = (\varphi_B^V)^{-1} L^* \varphi_C^W$: $[L^t]_{\mathfrak{C}}^{\mathfrak{B}} = [(\varphi_B^V)^{-1}]_{\mathfrak{B}^*}^{\mathfrak{B}} [L^*]_{\mathfrak{C}^*}^{\mathfrak{B}^*} [\varphi_C^W]_{\mathfrak{C}}^{\mathfrak{C}^*}$, with $[L^*] = ([L]_{\mathfrak{B}}^{\mathfrak{C}})^t$ (Observation 13.6.8) and $[\varphi_B^V] = J_{\mathfrak{B}}$, $[\varphi_C^W] = J_{\mathfrak{C}}$ (Observation 13.8.3).

(b) is a special case. ■

(This is the "secret" behind the transpose; compare Observation 13.1.8.)

Exercise 13.9.5

If $\mathfrak{B}, \mathfrak{C}$ are orthonormal bases, $[L^t]_{\mathfrak{C}}^{\mathfrak{B}} = ([L]_{\mathfrak{B}}^{\mathfrak{C}})^t$.

Solution. $J_{\mathfrak{B}} = J_{\mathfrak{C}} = I$ in Proposition 13.9.4. □

Exercise 13.9.6

(a) For $L(x_1, x_2, x_3, x_4)^t = (x_4, x_1, x_2, x_3)^t$ on $\mathbb{R}^4$ with the dot product, find L^t. (b) For $J = \mathrm{diag}(1, -1)$ and $L(x, y)^t = (x + y, -x + 3y)^t$ on $(\mathbb{R}^2, B_{\mathcal{E}}^J)$, find L^t.

Solution. (a) $[L]$ is a permutation matrix; $L^t = L^{-1}$: $L^t(x_1, x_2, x_3, x_4)^t = (x_2, x_3, x_4, x_1)^t$. (b) $[L] = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 3 \end{smallmatrix}\right)$ and $J^{-1}[L]^t J = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 3 \end{smallmatrix}\right)$: $L^t = L$ ("self-transpose" for this form, though $[L]$ is not a symmetric matrix). □

Proposition 13.9.7

$\mathbf{O}(V, B) = \{L \in \mathrm{GL}(V) : L^t \circ L = I\}$.

Proof. For $L \in \mathrm{GL}(V)$: $B(v, w) = B(Lv, Lw) = B(v, (L^t \circ L)w)$ for all v, w iff $L^t L = I$ (Observation 13.9.2). ■

Remark 13.9.8

From $A^t J A = J$ one might look for an operator M with $L^t M L = M$; this is a dead end, since J is the matrix of a bilinear form, not of an operator.

Observation 13.9.9

$\mathbf{t} : \mathfrak{L}(V, W) \to \mathfrak{L}(W, V)$, $L \mapsto L^t$, is an isomorphism; in particular $(L + M)^t = L^t + M^t$ and $(cL)^t = cL^t$.

Proof. It is the composite of the isomorphism $*$ (Exercise 13.6.10) with the identifications. ■

Observation 13.9.10

For $L \in \mathfrak{L}(V, W)$, $M \in \mathfrak{L}(W, U)$: (a) $I^t = I$;
(b) $(M \circ L)^t = L^t \circ M^t$;
(c) if L is invertible, so is L^t, with $(L^t)^{-1} = (L^{-1})^t$;
(d) $(L^t)^t = L$.

Proof. (a), (b) from Exercise 13.6.9. (c) from (a), (b).

(d) By Proposition 13.9.3 twice: $C(Lv, w) = B(v, L^t w) = C((L^t)^t v, w)$; apply Observation 13.9.2. ■

Exercise 13.9.11

Prove Observations 13.9.9 and 13.9.10(d) directly from $L^t = (\varphi_B^V)^{-1} L^* \varphi_C^W$.

Solution. **t** is linear because $*$ and composition with fixed isomorphisms are. For (d), $(L^t)^t = (\varphi_C^W)^{-1}(L^t)^*\varphi_B^V$ and $(L^t)^* = (\varphi_C^W)^* L^{**}((\varphi_B^V)^{-1})^*$. Using $(\varphi_B^V)^* = \varphi_B^V$ (Exercise 13.8.4, B symmetric) and $L^{**} = L$ (Exercise 13.6.12), this gives L. □

Exercise 13.9.12

For $L \in \mathfrak{L}(V,V)$: (a) $\mathrm{tr} L^t = \mathrm{tr} L$;
(b) $\det L^t = \det L$;
(c) $\phi_{L^t} = \phi_L$;
(d) $m_{L^t} = m_L$.

Solution. By Proposition 13.9.4(b), $[L^t] = J^{-1}[L]^t J$ is similar to $[L]^t$, which has the same trace, determinant and characteristic polynomial as $[L]$. For (d), $f(L^t) = f(L)^t$ for polynomials f (Observations 13.9.9, 13.9.10), so $f(L^t) = 0 \iff f(L) = 0$. □

Exercise 13.9.13

For $L \in \mathfrak{L}(V,W)$: (a) $\ker(L^t) = (\mathrm{im}\, L)^\perp$;
(b) $\mathrm{im}(L^t) = (\ker L)^\perp$.

Solution. (a) $L^t w = 0 \iff B(v, L^t w) = 0\ \forall v \iff C(Lv, w) = 0\ \forall v \iff w \in (\mathrm{im}\, L)^\perp$. (b) Apply (a) to L^t (using $(L^t)^t = L$) and take $\perp$: $\mathrm{im}\, L^t = ((\mathrm{im}\, L^t)^\perp)^\perp = (\ker L)^\perp$. □

Exercise 13.9.14

Explain the relation between Exercises 13.7.7 and 13.9.13 from the point of view of Proposition 13.8.10.

Solution. Under B- and C-identification $L^t = L^*$, $\perp$= perp (Proposition 13.8.10, Theorem 13.8.11); so Exercise 13.9.13 is Exercise 13.7.7 transported along these identifications. □